SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5 - ASSESSMENT TOOL 3 - Hackathon

Course Code:	**DSA0404**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO5 –Hackathon**	**Assessment:**	**Assessment Tool 3-Hackathon**
Weightage:	**40% of CIA**	**Total Marks:**	**100**
CO5 Statement:	Analyze and evaluate real-world challenges to design and implement innovative technology-based solutions using problem-solving, teamwork, and rapid prototyping techniques. **(BL4)**		
Student Name:	Y Sravan Kumar	**Reg. No.:**	192472289
Section / Batch:	CSE AI	**Date:**	31-08-2026

Code of Conduct

I __Y SRAVAN KUMAR(192472289)________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: SRAVAN Y

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

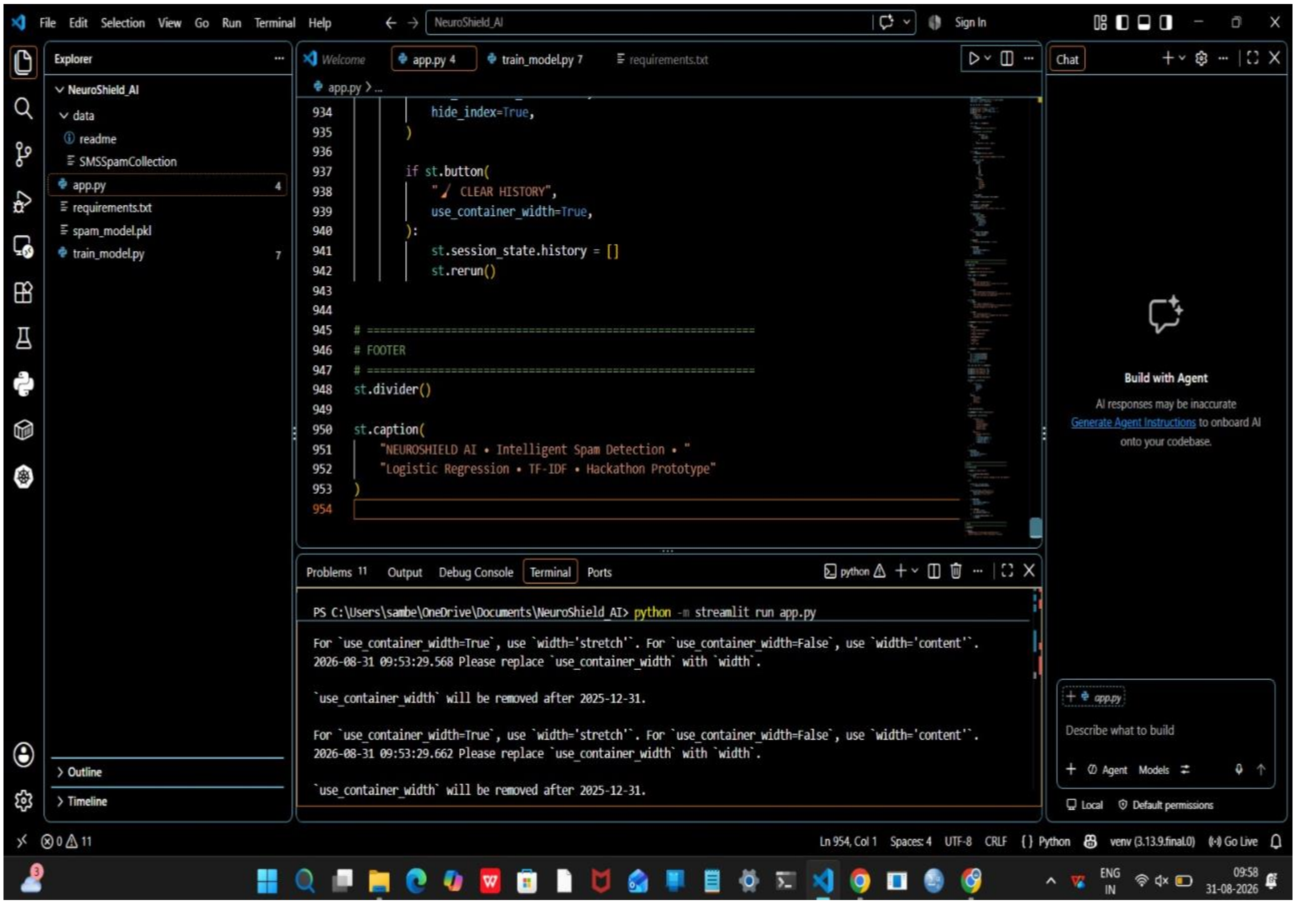
Problem Statement

With the rapid growth of digital communication, email and SMS have become common channels for sharing information. However, the increasing volume of unwanted and fraudulent messages, commonly known as spam, creates significant challenges for users. Spam messages may contain advertisements, misleading information, phishing links, or potentially harmful content, making the accurate identification of such messages important for secure and efficient communication.

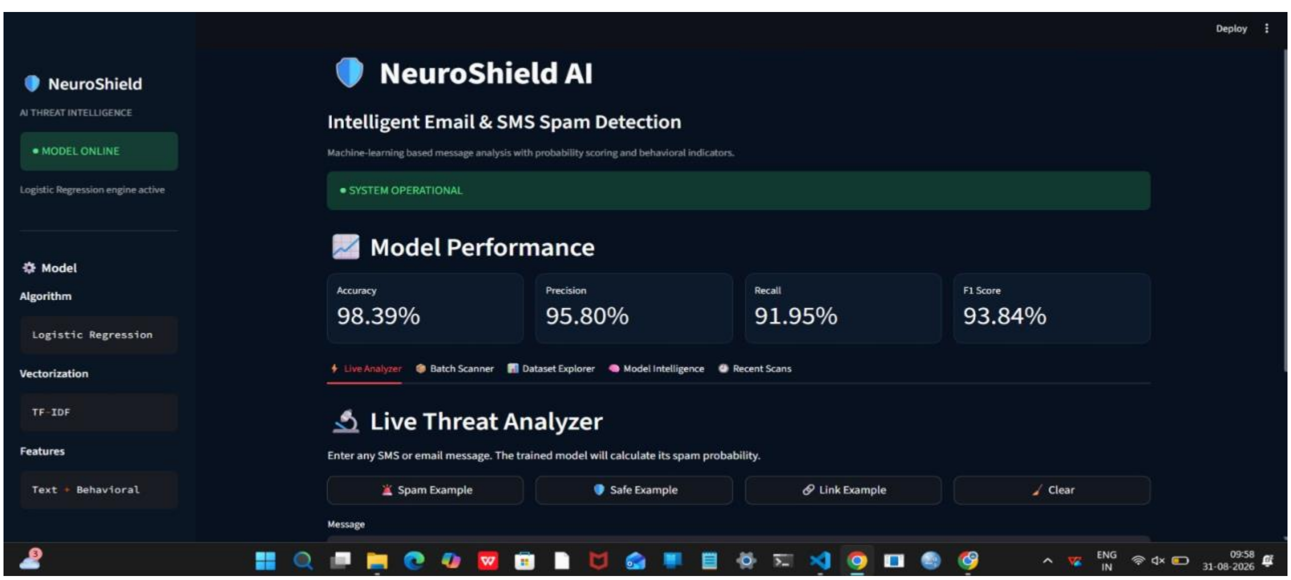
The problem addressed in this project is to develop an Email Spam Detection system that can automatically classify incoming emails or SMS messages as Spam or Not Spam (Ham). The system will use data science and machine learning techniques to analyse text-based features extracted from messages. Features such as word frequency, message length, presence of links, and other relevant textual characteristics will be considered for classification.

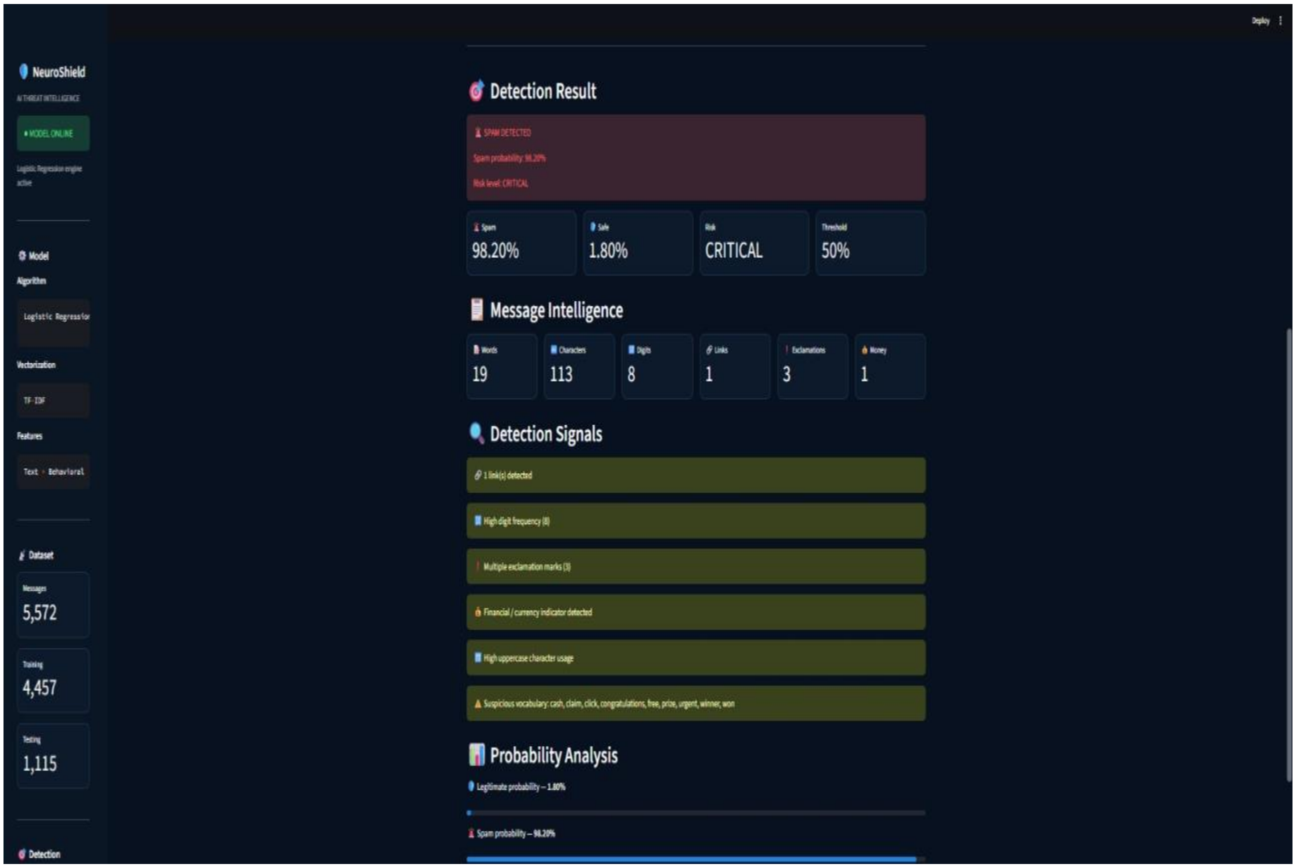
A Logistic Regression machine learning algorithm will be trained on a labelled dataset of spam and non-spam messages. The performance of the developed model will be evaluated using appropriate classification metrics, particularly precision, recall, and F1-score. The proposed system aims to provide an efficient and reliable method for detecting unwanted messages while minimizing incorrect classifications of legitimate messages.

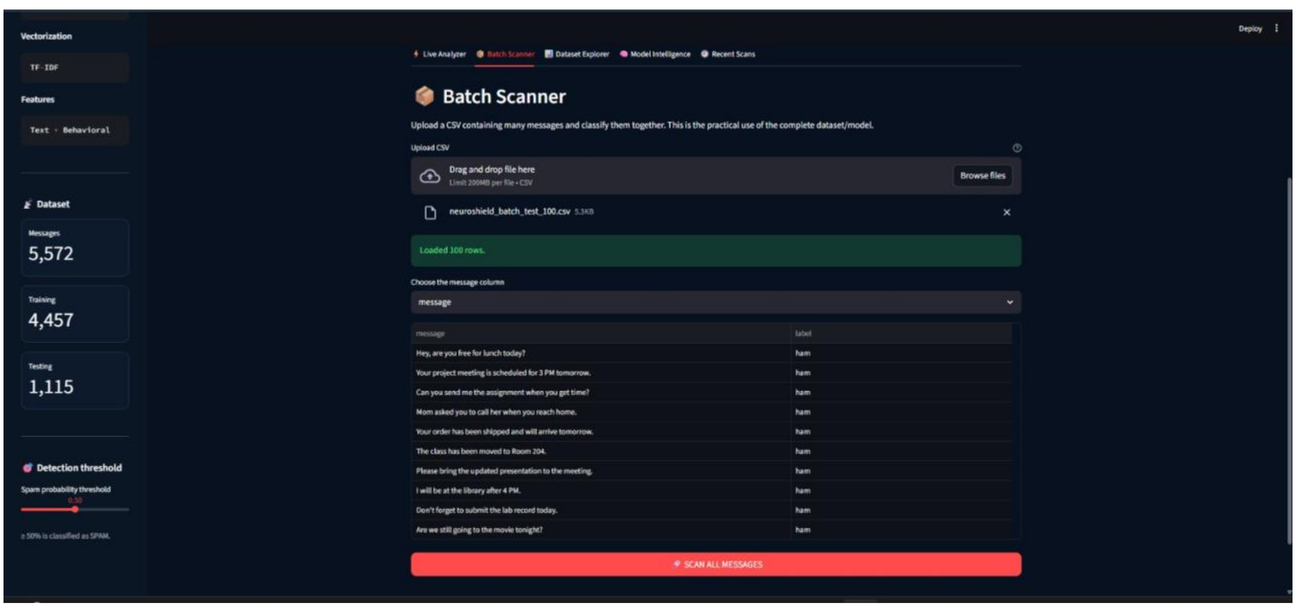
Implementation



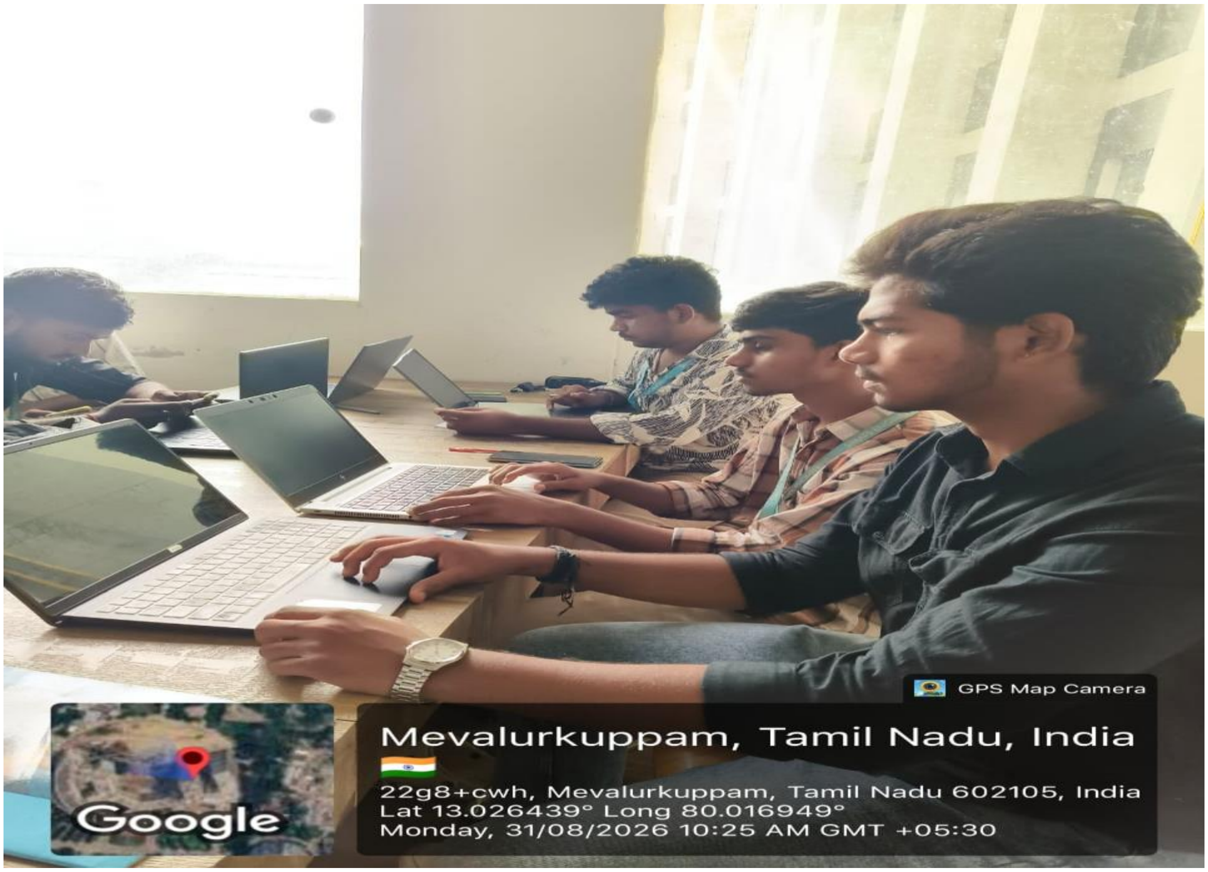
Output




NeuroShield
AI THREAT INTELLIGENCE
MODEL ONLINE
Logistic Regression engine active
Model
Algorithm
Logistic Regressio
Vectorization
TF-IDF
Features
Text + Behavioral
Dataset
Messages
5,572
Training
4,457
Testing
1,115
Detection
Deploy
Detection Result
SPAM DETECTED
Spam probability: 98.20%
Risk level: CRITICAL
Spam
98.20%
Safe
1.80%
Risk
CRITICAL
Threshold
50%
Message Intelligence
Words
19
Characters
113
Digits
8
Links
1
Exclamations
3
Money
1
Detection Signals
1 link(s) detected
High digit frequency (8)
Multiple exclamation marks (3)
Financial / currency indicator detected
High uppercase character usage
Suspicious vocabulary: cash, claim, click, congratulations, free, prize, urgent, winner, won
Probability Analysis
Legitimate probability — 1.80%
Spam probability — 98.20%


Vectorization
TF-IDF
Features
Text + Behavioral
Dataset
Messages
5,572
Training
4,457
Testing
1,115
Detection threshold
Spam probability threshold
0.50
≥ 50% is classified as SPAM.
Deploy
Live Analyzer
Batch Scanner
Dataset Explorer
Model Intelligence
Recent Scans
Batch Scanner
Upload a CSV containing many messages and classify them together. This is the practical use of the complete dataset/model.
Upload CSV
Drag and drop file here
Limit 200MB per file • CSV
Browse files
neuroshield_batch_test_100.csv 5.3KB
Loaded 100 rows.
Choose the message column
message
message
label
Hey, are you free for lunch today?
ham
Your project meeting is scheduled for 3 PM tomorrow.
ham
Can you send me the assignment when you get time?
ham
Mom asked you to call her when you reach home.
ham
Your order has been shipped and will arrive tomorrow.
ham
The class has been moved to Room 204.
ham
Please bring the updated presentation to the meeting.
ham
I will be at the library after 4 PM.
ham
Don't forget to submit the lab record today.
ham
Are we still going to the movie tonight?
ham
SCAN ALL MESSAGES

Geo Tag


GPS Map Camera
Mevalurkuppam, Tamil Nadu, India
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